# AarogyamCareAI

# Software Testing Documentation Portfolio

Prepared for final-year project submission | 20 October 2026

# Software Test Plan

# AarogyamCareAI Software Test Plan

Project: AarogyamCareAI — AI-Powered Healthcare Companion

Document status: Active

Prepared: 20 October 2026

## 1. Objectives and scope

This plan verifies that AarogyamCareAI provides reliable, authorized, and safe healthcare-information workflows without using real patient data. The scope covers the React/Vite user and administrator interfaces, the Node.js API, FastAPI machine-learning service, and MySQL-backed persistence.

In scope are authentication, role-based administration, chat history, disease and health-risk prediction, report analysis, appointments, and database-backed user isolation. Live Gemini and Twilio services are not load-tested; controlled mock responses are used for routine browser automation.

## 2. Test environment

| Component | Test environment |
| --- | --- |
| --- | --- |
| User UI | React/Vite, http://localhost:5173 |
| Admin UI | React/Vite admin mode, http://localhost:5174 |
| Node API | Node.js/Express, http://localhost:3000 |
| ML API | FastAPI, http://127.0.0.1:8000 |
| Persistence | Local MySQL, aarogyamcareai schema |
| Browser tests | Python 3.14.2, Selenium 4.51.0, pytest 9.1.1, pytest-html 4.2.0, Chrome |
| Performance tool | Apache JMeter 5.6.3 with Java 21 |

All test accounts, appointment data, questions, and report files must be synthetic. Credentials, API keys, database passwords, and real health data must never be committed or placed in reports.

## 3. Test strategy

| Category | Purpose | Primary tool | Status rule |
| --- | --- | --- | --- |
| --- | --- | --- | --- |
| Manual functional testing | Validate workflows and usability | Browser + manual suite | Mark only after manual execution |
| UI regression testing | Repeatable browser checks for core workflows | Selenium, pytest | Evidence from pytest result and screenshots |
| API testing | Validate status codes, schemas, authorization, and errors | Postman | Execute only against local controlled services |
| Integration testing | Verify UI, API, ML, and MySQL work together | Selenium + controlled local stack | Use synthetic data and mocks where needed |
| Database testing | Verify persistence, constraints, and relationships | MySQL Workbench | Prefer read-only verification or isolated records |
| Performance testing | Establish controlled response-time baselines | JMeter | Start with low load and increase only with approval |

| Category | Purpose | Primary tool | Status rule |
|---|---|---|---|
| Security testing | Verify authentication, authorization, validation, and secret hygiene | Source review + safe local requests | No brute force, destructive payloads, or uncontrolled scans |
| Regression testing | Detect functional regressions after changes | Full Selenium suite | Record actual pass/fail/skip counts |

## 4. Entry and exit criteria

Entry criteria: local services are running; MySQL is available; dedicated user and admin test accounts exist; test credentials are supplied through environment variables rather than source code.

Exit criteria: planned tests are executed or explicitly marked NOT EXECUTED; failures have evidence and a disposition; generated reports identify their actual execution date and environment.

## 5. Risks and controls

| Risk | Control |
|---|---|
| --- | --- |
| Real data disclosure | Use synthetic accounts/files; avoid raw health-record output in evidence |
| External-service cost or side effects | Mock Gemini; do not send test SMS; do not load-test third-party services |
| Test data left behind | Use unique test data and delete temporary chats/appointments where supported |
| Unreliable browser timing | Use Selenium explicit waits, not arbitrary sleeps |
| Regression during testing | Keep test code separate from application logic; make minimal test-only seams only when needed |
| Misleading results | Never report unexecuted tests as passed |

## 6. Documentation index

• Detailed manual cases: manual-functional-test-suite.md(manual-functional-test-suite.md)

• Selenium evidence: selenium-test-execution-report.md(selenium-test-execution-report.md)

• Resolved test-automation issues: defect-report.md(defect-report.md)

• Consolidated status: test-summary-report.md(test-summary-report.md)

# Manual Functional Test Suite

# AarogyamCareAI Manual Functional Test Suite

Document status: Ready for execution.

Execution status: No manual case in this document has been executed yet.

Data policy: Use the synthetic accounts and synthetic report files below; never enter real patient data, real telephone numbers, or production credentials.

## Test environment and data

Start the local stack from the repository root:

```
npm.cmd run dev:all
```

Expected services: user UI http://localhost:5173, admin UI http://localhost:5174, Node API http://localhost:3000, and ML API http://localhost:8000. MySQL must be available for database-backed flows.

Create these dedicated accounts before testing (or record their existing non-production equivalents):

| Alias | Role | Purpose |
| --- | --- | --- |
| TEST_USER_A | user | Primary functional-test account |
| TEST_USER_B | user | Data-isolation checks |
| TEST_ADMIN | admin | Admin-panel checks |

Suggested synthetic input: symptoms returned by /symptoms; risk values age 35, female, height 165, weight 62, systolic 120, diastolic 80, cholesterol 1, glucose 1, smoking false, alcohol false, physical activity true; appointment doctor Dr Test, a future date, time 10:00, reminder 30m.

Prepare harmless local files: a PNG, JPG, PDF, unsupported TXT, and a PDF/PNG larger than 10 MB. Do not use genuine medical reports.

## Recording rules

• Enter the observed browser/API/database evidence in Actual result while executing the case.

• Change Status only to PASS, FAIL, or BLOCKED after execution. Retain NOT EXECUTED otherwise.

• Record a defect ID beside a failed case. Severity guidance: Critical = security/data loss; High = core workflow unavailable; Medium = incorrect validation/persistence; Low = cosmetic or usability issue.

## Authentication and session handling

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| AUTH-01 | User login with valid credentials | Node API, MySQL, and TEST_USER_A available | 1. Open /login. 2. Enter valid email/password. 3. Select Login. | TEST_USER_A | Navigates to /dashboard; authenticated user session is active. | NOT EXECUTED | NOT EXECUTED | High |
| AUTH-02 | Reject invalid user login | Login page open | 1. Enter valid-format email with wrong password. 2. Select Login. | Wrong password | Stays on login page and shows an incorrect-credentials message; no session is created. | NOT EXECUTED | NOT EXECUTED | High |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| AUTH-03 | Validate empty login credentials | Login page open | 1. Leave fields blank. 2. Select Login. | Empty email/ password | Email and password validation messages display; no API login is attempted. | NOT EXECUTED | NOT EXECUTED | Medium |
| AUTH-04 | Logout ends user session | TEST_USER_A is logged in | 1. Use the dashboard logout control. 2. Attempt to open /dashboard in the same browser. | None | Returns to user login and protected dashboard is inaccessible without signing in. | NOT EXECUTED | NOT EXECUTED | High |
| AUTH-05 | Session survives permitted refresh and protects routes | TEST_USER_A is logged in | 1. Refresh /dashboard. 2. Open a protected route in a new tab. 3. Logout and repeat. | None | Session remains available before logout; protected routes redirect after logout. | NOT EXECUTED | NOT EXECUTED | High |

## Gemini chatbot and chat history

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| CHAT-01 | Send valid health-information question | TEST_USER_A logged in; Node API and configured Gemini key available | 1. Open /chat. 2. Enter a question. 3. Send. | What are common causes of a mild headache? | User message renders, loading indicator appears, and an informational assistant reply renders with the disclaimer. | NOT EXECUTED | NOT EXECUTED | High |
| CHAT-02 | Prevent empty message | Chat open | 1. Leave chat input blank/whitespace. 2. Try to send. | Empty text | Send control remains unavailable or no request is sent; no empty message renders. | NOT EXECUTED | NOT EXECUTED | Medium |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| CHAT-03 | Handle Gemini/API failure safely | Logged in; temporarily use unavailable AI configuration only in local test environment | 1. Send a valid question while the AI service is unavailable. 2. Observe message. | Valid question | A safe, understandable error appears; no token, stack trace, or secret is shown. | NOT EXECUTED | NOT EXECUTED | High |
| CHAT-04 | Verify conversation persistence and history | Complete CHAT-01 successfully | 1. Open /chat-history. 2. Find the conversation. 3. Open it. 4. Refresh. | CHAT-01 conversation | Conversation title/preview appears; original messages load after reopening and refresh. | NOT EXECUTED | NOT EXECUTED | High |
| CHAT-05 | Delete only selected conversation | At least two test conversations exist | 1. Delete one history entry. 2. Refresh. 3. Open the other entry. | Two synthetic chats | Deleted conversation is absent after refresh; other conversation is intact. | NOT EXECUTED | NOT EXECUTED | Medium |

## Disease prediction

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| DIS-01 | Obtain symptom list and make a valid prediction | Logged in; ML API and disease model available | 1. Open /disease-prediction. 2. Select one or more offered symptoms. 3. Predict. | Model-provided symptoms | A possible-condition result, numeric confidence, and educational disclaimer display. | NOT EXECUTED | NOT EXECUTED | High |
| DIS-02 | Validate no symptoms selected | Disease prediction page loaded | 1. Select no symptoms. 2. Try Predict. | None | Prediction action is disabled or validation prevents a request. | NOT EXECUTED | NOT EXECUTED | Medium |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| DIS-03 | Constrain invalid symptom input | Disease prediction page loaded | 1. Inspect available choices. 2. Try to select an unsupported symptom through the UI. | A symptom not in /symptoms | UI permits only model-supplied symptoms; unsupported free-text input cannot be submitted. Malformed payload validation is covered in API testing. | NOT EXECUTED | NOT EXECUTED | Medium |
| DIS-04 | Handle ML service failure | Logged in; controlled local ML outage | 1. Select valid symptoms. 2. Submit while ML API is unavailable. 3. Retry after recovery. | Valid selected symptoms | Clear availability error is shown without application crash or internal traceback; retry succeeds after recovery. | NOT EXECUTED | NOT EXECUTED | High |
| DIS-05 | Persist prediction history | Complete DIS-01 with Node API/MySQL available | 1. Make prediction. 2. Confirm through later API/database test or admin Predictions page. | DIS-01 result | One history record belongs to TEST_USER_A, with selected symptoms, result, confidence, and model name. | NOT EXECUTED | NOT EXECUTED | High |

## Health risk prediction

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| RISK-01 | Submit valid cardiovascular input | Logged in; ML API and risk model available | 1. Open /risk-prediction. 2. Complete every field. 3. Select Check Health Risk. | Suggested synthetic risk data | Displays Lower/Higher Risk, model name, score, and educational notice. | NOT EXECUTED | NOT EXECUTED | High |
| RISK-02 | Validate missing fields | Risk page open | 1. Leave all fields empty. 2. Select Check Health Risk. | Empty form | Field-specific required/selection errors display; no ML request is made. | NOT EXECUTED | NOT EXECUTED | Medium |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| RISK-03 | Validate invalid numeric and blood-pressure relation | Risk page open | 1. Enter age below 18 or height above 250. 2. Enter systolic less than/equal to diastolic. 3. Submit. | age=17; 120/120 BP | Range and blood-pressure validation messages display; no result is shown. | NOT EXECUTED | NOT EXECUTED | Medium |
| RISK-04 | Persist health-risk assessment | Complete RISK-01 with Node API/MySQL available | 1. Submit valid assessment. 2. Confirm through later API/database test or admin Predictions page. | RISK-01 result | One assessment record belongs to TEST_USER_A, with input snapshot, result, probability, and model name. | NOT EXECUTED | NOT EXECUTED | High |

## Health reports and report analysis

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| REP-01 | Upload supported PNG metadata | Logged in; Node API/MySQL available | 1. Open /reports. 2. Browse for PNG. 3. Upload. 4. Refresh. | Synthetic PNG | Selected filename is shown; report entry appears with upload status and persists after refresh. | NOT EXECUTED | NOT EXECUTED | High |
| REP-02 | Upload supported JPG and PDF metadata | Logged in | Repeat REP-01 once for JPG and once for PDF. | Synthetic JPG/PDF | Each file is accepted and appears as a distinct report record. | NOT EXECUTED | NOT EXECUTED | High |
| REP-03 | Reject unsupported type for analysis | Logged in | 1. Open /report-analysis. 2. Select TXT. 3. Analyze. | Synthetic TXT | Error states that only PDF, PNG, or JPG are accepted; no analysis starts. | NOT EXECUTED | NOT EXECUTED | Medium |
| REP-04 | Reject oversized report analysis input | Logged in | 1. Select supported file larger than 10 MB. 2. Analyze. | Synthetic >10 MB PDF/PNG | Error states the report must be smaller than 10 MB; no model request succeeds. | NOT EXECUTED | NOT EXECUTED | Medium |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| REP-05 | Analyze supported PNG/JPG/PDF | Logged in; Node API and Gemini configured | 1. Select each supported synthetic report. 2. Analyze. | Synthetic PNG, JPG, PDF | Results show Summary, Key Findings, Important Values, Possible Concerns, and Questions to Ask a Doctor, plus disclaimer. | NOT EXECUTED | NOT EXECUTED | High |
| REP-06 | Handle report-analysis API failure | Logged in; controlled local AI outage | 1. Select valid report. 2. Analyze during outage. | Valid synthetic report | Safe user-facing error displays; no raw base64 data, secret, or stack trace is exposed. | NOT EXECUTED | NOT EXECUTED | High |
| REP-07 | Verify report-analysis persistence | Complete REP-05 with MySQL available | 1. Navigate to admin Reports as TEST_ADMIN. 2. Locate report. 3. View analysis if provided. | REP-05 report | Metadata and analysis status /summary are present; raw uploaded file is not exposed in the admin UI. | NOT EXECUTED | NOT EXECUTED | High |

## Appointments and SMS consent

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| APT-01 | Create appointment without SMS consent | Logged in | 1. Open /appointments. 2. Fill doctor, future date/time, reminder. 3. Keep SMS consent off. 4. Save. | Suggested synthetic appointment | Appointment appears as upcoming; SMS status is Not scheduled; record persists after refresh. | NOT EXECUTED | NOT EXECUTED | High |
| APT-02 | Validate appointment required fields | Appointments page open | 1. Clear doctor /date/time as permitted. 2. Save. | Missing required values | Required-field error appears; no invalid record is created. | NOT EXECUTED | NOT EXECUTED | Medium |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| APT-03 | Enforce SMS consent and E.164 phone validation | Appointments page open | 1. Enable SMS consent without phone. 2. Save. 3. Enter malformed phone and retry. | Empty phone; 98765 | Explains phone is required with consent and requires international E.164 format. | NOT EXECUTED | NOT EXECUTED | Medium |
| APT-04 | Create SMS-consented appointment safely | Twilio configuration status known; use a permitted synthetic/test recipient only | 1. Enter valid E.164 test number. 2. Consent to SMS. 3. Save. | Authorized test number only | Appointment is scheduled or clearly reports service unavailability; no SMS is sent without explicit consent. | NOT EXECUTED | NOT EXECUTED | High |
| APT-05 | Edit, cancel, and verify persistence | Complete APT-01 | 1. Edit doctor/time and save. 2. Refresh. 3. Delete. 4. Refresh. | APT-01 item | Edit survives refresh; delete removes the item from the current user only. | NOT EXECUTED | NOT EXECUTED | High |

## Admin panel and role restrictions

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| ADM-01 | Admin login and dashboard | TEST_ADMIN exists; Node/MySQL available | 1. Open /admin/login. 2. Sign in with admin account. | TEST_ADMIN | Opens /admin; dashboard displays database-backed metrics or a safe availability error. | NOT EXECUTED | NOT EXECUTED | High |
| ADM-02 | Monitor users and appointments | Admin logged in; test data exists | 1. Open Users. 2. Search/filter role and paginate if applicable. 3. Open Appointments. | Test accounts/appointments | User and appointment information renders; password hashes and patient phone numbers are not displayed. | NOT EXECUTED | NOT EXECUTED | High |

| ID | Module / objective | Preconditions | Steps | Test data | Expected result | Actual result | Status | Severity if failed |
|---|---|---|---|---|---|---|---|---|
| ADM-03 | Monitor reports and ML predictions | Admin logged in; REP-05/DIS-01/RISK-01 data exists | 1. Open Reports. 2. View available analysis. 3. Open ML Predictions. | Prior synthetic records | Only metadata/analysis output and stored prediction summaries appear; raw files and model-management controls are absent. | NOT EXECUTED | NOT EXECUTED | High |
| ADM-04 | Check SMS UI and test function | Admin logged in; Twilio status known | 1. Open SMS & Reminders. 2. Confirm configuration state. 3. Use only an authorized test recipient if sending. | Authorized test recipient | SMS UI is admin-only; validation rejects invalid phone/message; result records success/failure safely. | NOT EXECUTED | NOT EXECUTED | High |
| ADM-05 | Prevent normal-user admin access | TEST_USER_A logged in | 1. Browse to /admin. 2. Try each admin navigation URL. 3. Call no admin actions. | Normal user | UI redirects to dashboard and direct protected API calls later return 403, not data. | NOT EXECUTED | NOT EXECUTED | Critical |

## Execution log

| Date/time | Tester | Test IDs run | Build/commit | Browser | Passed | Failed | Blocked | Notes / defect IDs |
|---|---|---|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| NOT EXECUTED | | | | | 0 | 0 | 0 | |

# Selenium Test Execution Report

# Selenium Test Execution Report

Project: AarogyamCareAI

Execution date: 20 October 2026

Result: PASS

## Environment and command

The controlled local stack was started with npm run dev:test. It runs the user UI on port 5173, administrator UI on port 5174, Node API on port 3000, and FastAPI ML service on port 8000. Gemini chat and report analysis used deterministic local mock responses for repeatable automation.

```
.\tests\.venv\Scripts\python.exe -m pytest tests\selenium -v
--html="tests\reports\selenium-report-20261020-164529.html" --self-contained-html
```

## Actual result

| Metric | Result |
| --- | ---: |
| Collected tests | 20 |
| Passed | 20 |
| Failed | 0 |
| Skipped | 0 |
| Duration | 372.31 seconds (6 minutes 12 seconds) |

## Coverage achieved

| Module | Tests | Verified behavior |
| --- | ---: | --- |
| Landing-page smoke | 1 | Application-specific landing content renders |
| Authentication | 4 | Empty/invalid login validation, valid synthetic login, logout route protection |
| Chatbot | 3 | Empty-message prevention, mocked message exchange, conversation-history persistence |
| Disease prediction | 2 | Symptom validation and real local ML result rendering |
| Health-risk prediction | 3 | Required fields, blood-pressure rule, real local ML result rendering |
| Report analysis | 2 | File-required validation and mocked structured analysis rendering |
| Appointments | 2 | Validation, create, refresh persistence, edit, deletion, deletion persistence |
| Administration | 3 | User restriction, role-specific login rejection, administrator navigation, SMS-control visibility, logout |

## Evidence

The self-contained report is available at:

• selenium-report-20261020-164529.html(../reports/selenium-report-20261020-164529.html)

Failure screenshots are configured to be captured automatically during future runs. This execution had no failures, so no failure screenshot was generated.

## Limitations

The chat and report-analysis assertions intentionally use local deterministic mock responses. They do not certify live Gemini availability or behavior. No real SMS message was sent.

# Defect Report

# Defect Report

Project: AarogyamCareAI

Scope: Defects observed during the completed Selenium automation work

Application defects confirmed: None in the executed 20-test Selenium suite.

The issues below were test-automation implementation defects, not confirmed production-application defects.

| ID | Type | Severity | Summary | Evidence and resolution | Retest |
|---|---|---|---|---|---|
| --- | --- | --- | --- | --- | --- |
| TST-001 | Test automation | Medium | Appointment Page Object produced an invalid XPath for the label Doctor's name. | The apostrophe made the generated single-quoted XPath invalid. The locator was corrected to use a compatible XPath literal and appointment-card targeting was made specific. | Resolved: both appointment tests passed. |
| TST-002 | Test automation | Low | Admin test checked the SMS control after navigating away from the SMS page. | The assertion timed out because the test was on the System page. The test now verifies control visibility on the SMS page, then navigates to System. | Resolved: all three admin tests passed. |

## Defect handling rule

For future product defects, record reproduction steps, expected and actual outcomes, evidence, severity, proposed minimal fix, and retest status. Obtain approval before modifying production application logic.

# Test Summary Report

# AarogyamCareAI Test Summary Report

Report date: 20 October 2026

Assessment scope: Completed Selenium automation and its supporting local environment.

## Executive summary

The completed Selenium regression suite passed all 20 automated browser tests. The verified workflows include user authentication, mock-controlled chat and report analysis, local ML predictions, appointments, and administrator role restrictions. No application defect was confirmed during that executed scope.

## Execution totals

| Outcome | Count |
| --- | ---: |
| Automated tests executed | 20 |
| Passed | 20 |
| Failed | 0 |
| Skipped | 0 |
| Open application defects from this execution | 0 |

## Quality assessment

The executed browser-regression scope is suitable as a repeatable local baseline. It demonstrates successful interaction between React, Node/Express, FastAPI ML services, and MySQL for the covered workflows.

## Exclusions

At the user's request, no formal API, JMeter performance, MySQL database, or security execution report is included in this documentation package. Any work not represented by its own execution report must be treated as incomplete or NOT EXECUTED for submission purposes.

## Supporting documents

• Software Test Plan(software-test-plan.md)

• Manual Functional Test Suite(manual-functional-test-suite.md)

• Selenium Test Execution Report(selenium-test-execution-report.md)

• Defect Report(defect-report.md)